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Roll No.

220953

5th Sem/ Electrical, IC
Subject : Industrial Electronics & Control of Drives

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 SCR is

- a) Four layer, four junction device
- b) Four layer, three junction device
- c) Four layer, two junction device
- d) Three layer, single junction device

Q.2 DIAC is a

- a) Unidirectional device
- b) Bi directional device
- c) High power device
- d) All of the above

Q.3 Inverters converts

- a) DC power of DC power
- b) DC power to AC power
- c) AC power to AC power
- d) AC power to DC power

- Q.4 In a single phase half-wave circuit with RL load and a freewheeling diode, the load voltage during the freewheeling period will be
- a) Zero
 - b) Positive
 - c) Negative
 - d) Positive than negative
- Q.5 A single phase full controlled bridge converter uses
- a) 4 SCRs and 2 diodes
 - b) 4 SCRs
 - c) 6 SCRs
 - d) 4 SCRs and 2 diodes
- Q.6 Latching current is related to
- a) Turn-on process of SCR
 - b) Turn-off process of SCR
 - c) Both option 'a' and 'b'
 - d) None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 What is a cyclo converter?
- Q.8 Write full form of UPS.
- Q.9 What is inverter?
- Q.10 What is a Chopper?
- Q.11 What is a controlled rectifier?
- Q.12 What is an AC drive?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Describe application of TRIAC as fan regulator.
- Q.14 Write two methods of triggering of SCR.
- Q.15 Describe single phase full controlled full wave rectifier with Resistive load along with proper circuit diagram and voltages -Current wave forms)
- Q.16 Draw three phase full wave half controlled bridge rectifier.
- Q.17 Explain working of Parallel inverter.
- Q.18 Write working principle and two applications of choppers.
- Q.19 Differentiate between the ON line and OFF line UPS.
- Q.20 Explain concept of Cyclo converter controlled AC drive.
- Q.21 Describe working principle of SMPS?
- Q.22 Explain working of power converter for EV charging.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 What is Dual converter? Explain working principle and applications.

Q.24 Explain construction, working principle and V-I characteristics of SCR.

Q.25 Explain working of UJT based relaxation oscillator.